

INSTITUTE OF ACTUARIES OF INDIA

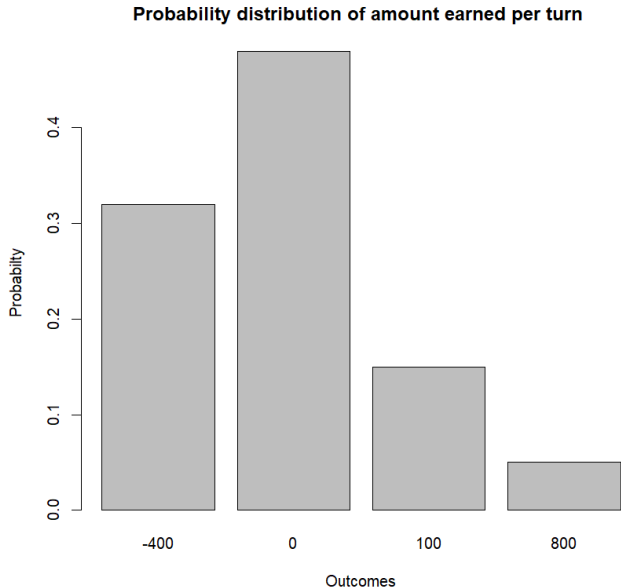
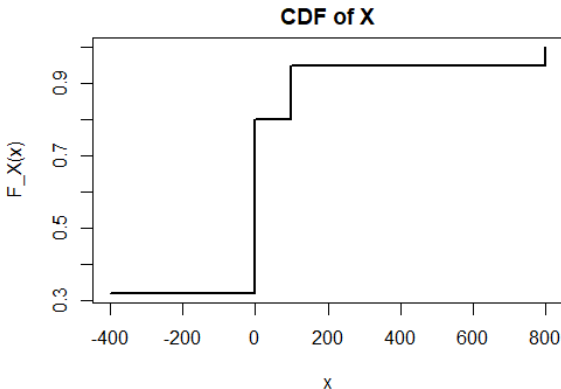
CS1 - Actuarial Statistics (Paper B)

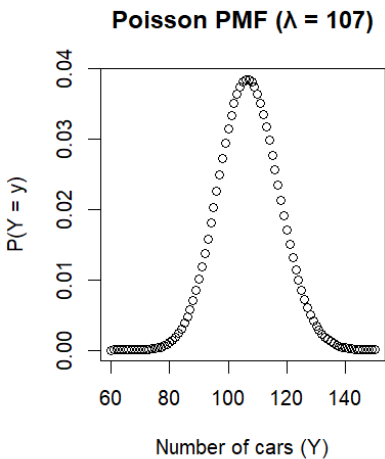
November 2025 Examination

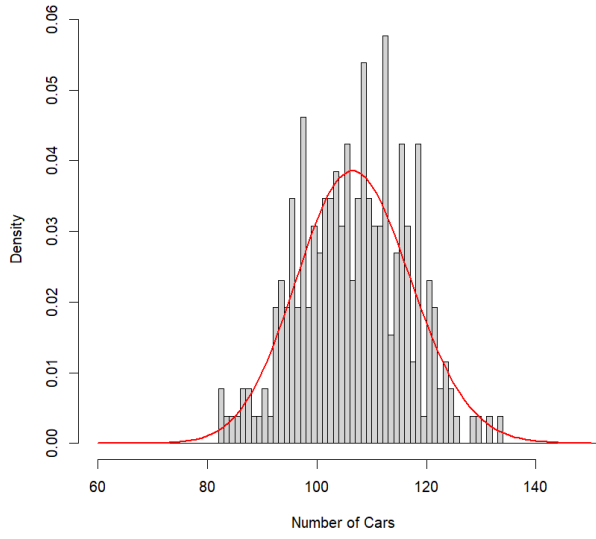
Indicative Solution

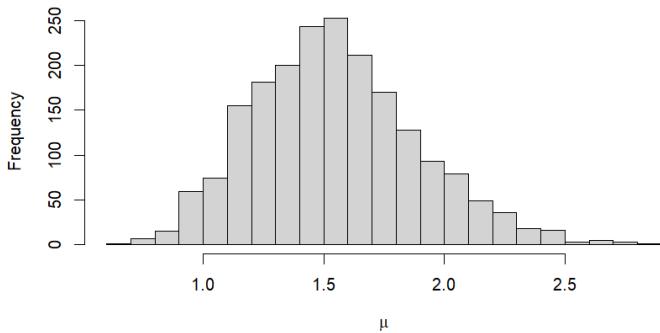
Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

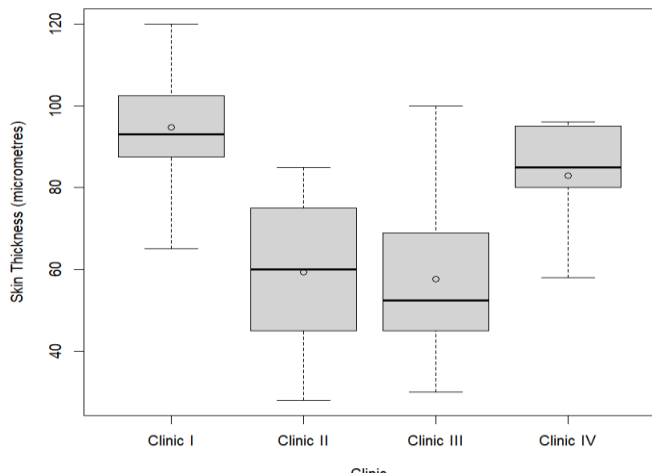
Solution 1:		
i)	<pre>outcomes <- c(-400, 0, 100, 800) prob <- c(0.32, 0.48, 0.15, 0.05)</pre>	(2)
ii)	<pre>barplot(prob,,names.arg=outcomes,xlab="Outcomes",ylab="Probabilty",main="Probability distribution of amount earned per turn ")</pre>  Probability distribution of amount earned per turn Outcomes	(2)
iii)	<pre>cumul<-cumsum(prob) print(cumul) 0.32 0.80 0.95 1.00</pre>	(2)
iv)	<pre>plot(outcomes, cumul, type = "s", lwd = 2, xlab = "x", ylab = " Pr(X<= x)", main = "CDF of X")</pre>  CDF of X F_X(x) x	(2)
v)	<pre>mean <- sum(outcomes *prob) mean -73 sd <- sqrt(sum((outcomes - mean)^2 * prob)) sd 281.7286</pre> alternate solution solution for sd is:	

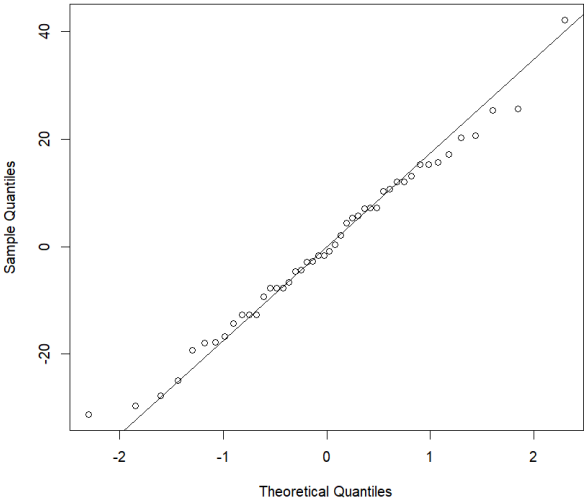
	<pre>sd<-sqrt(sum(outcomes^2*prob)-mean^2) sd</pre>	(3)
vi)	<pre>lambda <- 107 prob_more_than_100 <- 1-ppois(100, lambda) prob_more_than_100 0.7319128</pre>	(2)
vii)	<pre>prob_zero_cars <- dpois(0, lambda) prob_zero_cars 3.39227e-47</pre>	(1)
viii)	<pre>plot(y <- 60:150 , dpois(y, lambda), main = "Poisson PMF ($\lambda = 107$)", xlab = "Number of cars (Y)", ylab = "P(Y = y)")</pre> 	(2)
ix)	<pre>set.seed(2025) simulated <- rpois(260, lambda)</pre>	(1)
x)	<pre>hist(simulated, breaks = 60:151, xlim = c(60, 150), main = "Simulated vs Theoretical PMF of Cars Passing", xlab = "Number of Cars",freq=FALSE) lines(y, dpois(y,lambda), col = "red", lwd = 2)</pre>	

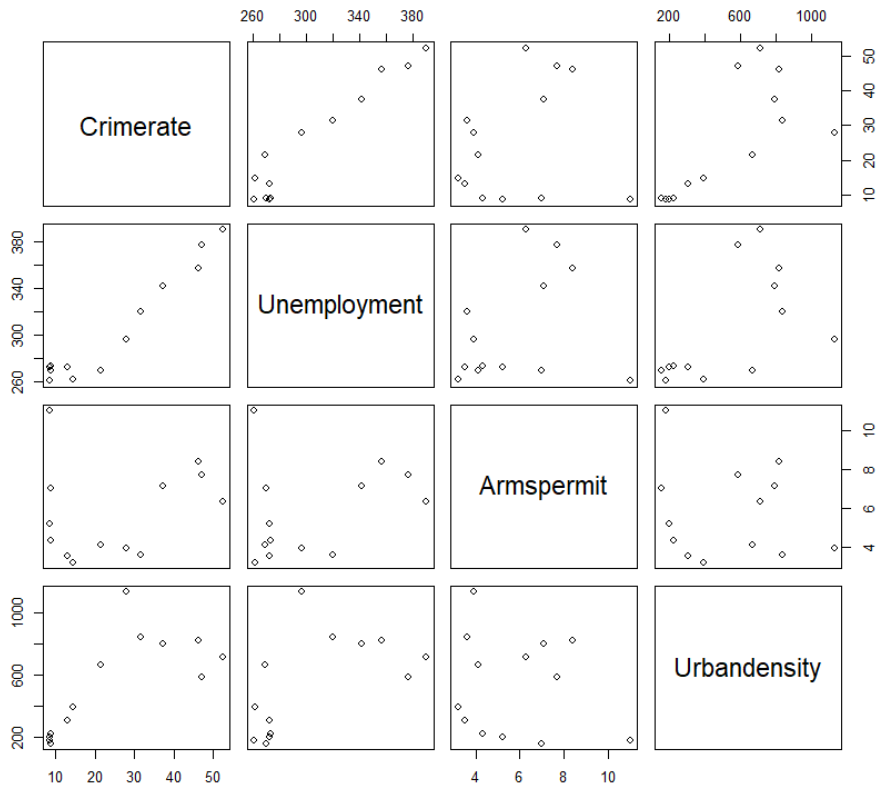
	Simulated vs Theoretical PMF of Cars Passing 	(3)
xi)	The histogram of the simulated data closely follows the shape of the theoretical PMF, confirming that the simulations are consistent with the Poisson model. Minor deviations are due to randomness and sample size.	(1)
	(21 marks)	
Solution 2:		
i)	<pre>X<- c(1,6,0,3,5,2,1,0)</pre> <pre>alpha_post <- 2 + sum(X)</pre> <pre>alpha_post</pre> <pre>20</pre> <pre>lamda_post <- lamda_prior + length(X)</pre> <pre>lamdba_post</pre> <pre>13</pre> Bayesian estimate for μ a. Squared Error loss = mean of gamma dist <pre>alpha_post / lamda_post</pre> <pre>1.538462</pre> b. All or nothing loss = mode of gamma dist <pre>(alpha_post -1) / lamda_post</pre> <pre>1.461538</pre>	(3)
ii)	<pre>qgamma(c(0.025, 0.975), shape = 20, rate = 13)</pre> <pre>0.9397323 2.2823734</pre> $W = \mu x \sim \text{Gamma}(20,13) \Leftrightarrow 2 \times 13W \sim \chi^2_{2 \times 20} = 26W \sim \chi^2_{40}$ 95% credible interval is given by <pre>qchisq(c(0.025,0.975), df = 40)/26</pre> <pre>0.9397323 2.2823734</pre>	(2)

iii)	<pre>set.seed(1234) posterior_samples <- rgamma(2000, shape = alpha_post, rate = lambda_post) tail(posterior_samples) 1.975092 1.470546 1.177162 1.670617 1.489956 2.268623</pre>	(2)																				
iv)	<pre>hist(posterior_samples, breaks = 30, main = "Posterior Distribution of μ", xlab = expression(mu))</pre> Posterior Distribution of μ 	(2)																				
v)	$\mu = 1$ - Lies within the 95% credible interval. - Close to the posterior mean/median/mode. - Reasonable value for μ based on the observed data. $\mu = 5$ - Lies far outside the 95% credible interval - Highly unlikely given the posterior distribution. - Strongly unfavoured based on the data and prior.	(3)																				
(12 Marks)																						
Solution 3:																						
i)	<pre>treatment_matrix <- matrix(c(20, 9, 16, 32, 72, 64, 8, 8, 30, 52, 32, 12), nrow = 4, byrow = TRUE) rownames(treatment_matrix) <- c("Injection", "Tablet", "Laser", "Herbal") colnames(treatment_matrix) <- c("None", "Partial", "Full") print(treatment_matrix)</pre> <table><thead><tr><th></th><th>None</th><th>Partial</th><th>Full</th></tr></thead><tbody><tr><td>Injection</td><td>20</td><td>9</td><td>16</td></tr><tr><td>Tablet</td><td>32</td><td>72</td><td>64</td></tr><tr><td>Laser</td><td>8</td><td>8</td><td>30</td></tr><tr><td>Herbal</td><td>52</td><td>32</td><td>12</td></tr></tbody></table>		None	Partial	Full	Injection	20	9	16	Tablet	32	72	64	Laser	8	8	30	Herbal	52	32	12	(3)
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Injection	20	9	16																			
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Laser	8	8	30																			
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ii)	H0: Success is independent of treatment method vs H1: Success depends on treatment method <pre>chisq.test(treatment_matrix)</pre> Pearson's Chi-squared test data: treatment_matrix X-squared = 66.166, df = 6, p-value = 2.492e-12																					

	p-value of the test is $2.492e-12$ Since $p\text{-value} < 0.05$, we reject the null hypothesis and.... ...conclude that there is a dependency between the treatment and level of success of the treatment	(4)
iii)	$\text{pnorm}(81.5, 80.2, 1.1) - \text{pnorm}(80.5, 80.2, 1.1)$ 0.2738925	(1.5)
iv)	$\text{qnorm}(0.2, 80.2, 1.1)$ 79.27422	(1.5)
v)	$\text{weights1} <- c(87.7, 80.01, 77.28, 78.76, 81.52, 74.2, 80.71, 79.5, 77.87, 81.94, 80.7, 82.32, 75.78, 80.19, 83.91, 79.4, 77.52, 77.62, 81.4, 74.89, 82.95, 73.59, 77.92, 77.18, 79.83, 81.23, 79.28, 78.44, 79.01, 80.47, 76.23, 78.89, 77.14, 69.94, 78.54, 79.7, 82.45, 77.29, 75.52, 77.21)$ $H_0: \mu = 80$ vs $H_1: \mu < 80$ $\text{t.test}(\text{weights1}, \text{mu} = 80, \text{alternative} = \text{"less"})$ One Sample t-test data: weights1 $t = -2.2268$, $df = 39$, $p\text{-value} = 0.0159$ alternative hypothesis: true mean is less than 80 95 percent confidence interval: -Inf 79.7325 sample estimates: mean of x 78.90075 p-value of the test is 0.0159 Since $p\text{-value} < 0.05$, we reject the null hypothesis and.... ...conclude that the mean of the sample packs is less than 80grams	(4)
vi)	Yes, since p-value is greater than 0.01, we fail to reject the null hypothesis and the inference will change implying that the mean weight of the packets is not less than 80grams	(1)
vii)	$\text{weights2} <- c(80.22, 79.73, 81.1, 78.76, 82.03, 81.66, 80.97, 81.32, 80.12, 78.98, 79.21, 81.48, 79.86, 81.06, 77.96, 80.73, 80.34, 80.01, 81.82, 79.3, 79.08, 79.47, 78.98, 80.87, 82.24, 77.22, 80.03, 79.2, 80.95, 79.17)$ $\text{mean}(\text{weights2})$ 80.129 The mean weight of the second sample 80.1 grams is higher than the mean weight of the first sample 78.9 grams and closer to expected mean of 80 grams	(2)
viii)	$H_0: \mu_2 = \mu_1$; $H_1: \mu_2 > \mu_1$ $\text{t.test}(\text{weights2}, \text{weights1}, \text{alternative} = \text{"greater"})$ welch Two Sample t-test	

	<pre>data: weights2 and weights1 t = 2.2666, df = 53.598, p-value = 0.01374 alternative hypothesis: true difference in means is greater than 0 95 percent confidence interval: 0.3212413 Inf sample estimates: mean of x mean of y 80.12900 78.90075</pre> p-value of the test is 0.01374 Since $p\text{-value} < 0.05$, we reject the null hypothesis and.... ...conclude that the mean weight of the second sample packs is greater than mean weight of the first sample	(4)
ix)	<pre>clinic1 <- c(93,120,65,105,115,82,99,87,100,90,78,95,93,88,110) clinic2 <- c(85,45,80,28,75,70,65,55,50,40) clinic3 <- c(100,75,65,40,73,65,50,30,45,50,45,55) clinic4 <- c(96,58,95,90,65,80,85,95,82)</pre> <pre>thickness <- c(clinic1, clinic2, clinic3, clinic4) clinic <- c(rep("Clinic I", length(clinic1)), rep("Clinic II", length(clinic2)), rep("Clinic III", length(clinic3)), rep("Clinic IV", length(clinic4)))</pre>	(3)
x)	<pre>boxplot(thickness ~ clinic, main = "Skin Thickness by Clinic", xlab = "Clinic", ylab = "Skin Thickness (micrometres)") means <- tapply(thickness, clinic, mean) points(1:4, means)</pre> Skin Thickness by Clinic 	(5)
xi)	<pre>a) residuals <- thickness - means[as.character(clinic)]</pre> <pre>qqnorm (residuals) qqline(residuals)</pre>	

	Normal Q-Q Plot  Since the sample quantiles are close to the normal quantiles, we can assume normality of the data b) <code>sds<- tapply(thickness,clinic,sd)</code> <pre>sds Clinic I Clinic II Clinic III Clinic IV 14.37591 18.69076 18.96947 13.55032</pre> <pre>max(sds)/min(sds) 1.399928</pre> Since, the ratio of the largest sample standard deviation to the smallest is less than 2, we can assume equality of variances.	(5)
		(3)
xii)	H0: All group means are equal vs H1: All group means are not equal <pre>anova_result <- aov(thickness ~ clinic) summary(anova_result)</pre> <pre> Df Sum Sq Mean Sq F value Pr(>F) clinic 3 12397 4132 15.14 7.99e-07 *** Residuals 42 11465 273 --- Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1</pre> p-value - 7.99e-07 Since p-values < 0.05, we reject the null hypothesis and Conclude that the mean skin thickness between clinics is significantly different.	(5)
	(42 marks)	
<u>Solution 4:</u>		

i)	a) <code>crime_data <- read.csv("C: \\planning.csv")</code> b) <code>plot(crime_data, main = "Scatterplot Matrix of Crime Data")</code> Scatterplot Matrix of Crime Data  c) <code>cor(crime_data\$Crimerate,crime_data\$Unemployment)</code> 0.964061 <code>cor(crime_data\$Crimerate,crime_data\$Armspermit)</code> 0.2101417 <code>cor(crime_data\$Crimerate,crime_data\$Urbandensity)</code> 0.7263054 d) Unemployment appears to be most strongly related to the crime rate	(1) (2) (3) (1)																																			
ii)	a) <code>model1<-lm(Crimerate ~ Unemployment + Armspermit + Urbandensity, data = crime_data)</code> <code>summary(model1)</code> Call: <code>lm(formula = Crimerate ~ Unemployment + Armspermit + Urbandensity, data = crime_data)</code> Residuals: <table><tr><td>Min</td><td>1Q</td><td>Median</td><td>3Q</td><td>Max</td></tr><tr><td>-2.8422</td><td>-1.9451</td><td>0.2012</td><td>0.9172</td><td>4.6694</td></tr></table> Coefficients: <table><tr><td></td><td>Estimate</td><td>Std. Error</td><td>t value</td><td>Pr(> t)</td></tr><tr><td>(Intercept)</td><td></td><td></td><td></td><td></td></tr><tr><td>Unemployment</td><td></td><td></td><td></td><td></td></tr><tr><td>Armspermit</td><td></td><td></td><td></td><td></td></tr><tr><td>Urbandensity</td><td></td><td></td><td></td><td></td></tr></table>	Min	1Q	Median	3Q	Max	-2.8422	-1.9451	0.2012	0.9172	4.6694		Estimate	Std. Error	t value	Pr(> t)	(Intercept)					Unemployment					Armspermit					Urbandensity					(2)
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	<pre>(Intercept) -68.852509 5.862631 -11.744 9.25e-07 *** crime_data\$Unemployment 0.280799 0.024657 11.388 1.20e-06 *** crime_data\$Armspermit 0.147248 0.408235 0.361 0.72665 crime_data\$Urbandensity 0.014177 0.003538 4.007 0.00308 ** ---</pre> Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1 Residual standard error: 2.89 on 9 degrees of freedom Multiple R-squared: 0.9767, Adjusted R-squared: 0.9689 F-statistic: 125.6 on 3 and 9 DF, p-value: 1.158e-07 b) Crimerate = -68.8525 + 0.2808 * Unemployment + 0.1472 * Armspermit + 0.0142 * Urbandensity Interpretation of coefficients: - o Intercept ≈ -68.85: model baseline (not directly meaningful alone). - o Unemployment: Each additional unemployed person per 100k is associated with ≈ 0.2808 more crimerate per 100k - o Armspermit: Each 1 percentage point increase in issued arms permit associated with ≈ 0.1472 more murders per 100k - o Urbandensity: Each additional resident per square kilometer in urban area is associated with ≈ 0.0142 more crimerate per 100k c) R^2 value of the model is 0.9767 which explain the amount of variation in the response attributed to the joint effect of the three variables as predictors p- value for Unemployment is 1.201109e-06 which is < 0.05 hence significant at 5% significance level Armspermit is 0.72655 which is > 0.05 hence nonsignificant Urbandensity is 0.00308 which is < 0.05 hence significant	(3)
iii)	a) Dropping Armspermit since it is the most insignificant parameter. <pre>model2 <- lm (Crimerate ~ Unemployment + Urbandensity, data=crime_data) > summary(model2)</pre> Call: lm(formula = Crimerate ~ Unemployment + Urbandensity, data=crime_data) Residuals: <pre> Min 1Q Median 3Q Max -2.9424 -2.1068 0.2775 0.9614 4.6408</pre> Coefficients: <pre> Estimate Std. Error t value Pr(> t) (Intercept) -69.002919 5.587647 -12.349 2.23e-07 *** crime_data\$Unemployment 0.285048 0.020697 13.772 7.92e-08 *** crime_data\$Urbandensity 0.013636 0.003062 4.453 0.00123 ** ---</pre> Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1 Residual standard error: 2.761 on 10 degrees of freedom Multiple R-squared: 0.9763, Adjusted R-squared: 0.9716 F-statistic: 206.3 on 2 and 10 DF, p-value: 7.417e-09	(3)

	Crimerate = $-69.003 + 0.2850 * \text{Unemployment} + 0.0136 * \text{Urbandensity}$ b) The new coefficient of determination is 0.9763 which is slightly lower than the previous one. However, there is an increase in the adjusted R-Squared value. The model is better off when Armspermit is not considered.	(2)						
iv)	a) <code>new_data_1 <- data.frame(Unemployment = 300, Urbandensity = 500)</code> <code>predict(model2, newdata = new_data_1, interval = "confidence", level = 0.99)</code> <table border="1"> <thead> <tr> <th>fit</th> <th>lwr</th> <th>upr</th> </tr> </thead> <tbody> <tr> <td>23.32948</td> <td>20.88251</td> <td>25.77645</td> </tr> </tbody> </table> 99% CI for is [20.88, 25.77] b) <code>coefs <- coef(model2)</code> $(0 - \text{coefs}[1] - \text{coefs}["\text{Urbandensity}] * 500) / \text{coefs}["\text{Unemployment}"]$ 218.1561 <i>Alternative solution: $(0 + 69,003 - 0.0136 * 500) / 0.2850 = 218.1561$</i> The number of unemployed people per 100,000 residents should be 218.16 for a mean crime rate of 0.	fit	lwr	upr	23.32948	20.88251	25.77645	(3) (2)
fit	lwr	upr						
23.32948	20.88251	25.77645						
		(25 marks)						
